

(22) Tropical Cyclone

Condition favorable for the formation of tropical cyclone

1. A large area of water surface with temperatures above 27 degrees Celsius
2. Coriolis force should be strong enough to form the cyclonic vortex.
3. A weak low pressure area or low-level cyclonic circulation which already exists on the ocean surface can reinforce and turn into a tropical cyclone.
4. There should be divergence around the 9 to 15 km. which sucks the air from the ocean surface above and thus the low pressure at the surface is strengthened.
5. Variation in the vertical wind speed is minor, this is referred to as low wind shear.
6. The tropical cyclone developed around the ITCZ

Favourable Conditions for the Formation of Tropical Cyclone

- A large area of water surface with a temperature above 27° C.
- The Coriolis force is strong enough to form a cyclonic vortex.
- A weak low-pressure area or low-level cyclonic circulation already exists on ocean surface
- There should be a divergence around 9 km to 15 km which sucks the air from ocean surface above and thus the upward movement of air is accelerated and low pressure centre at the surface is further intensified.
- Variations in the vertical wind speed are minor. This is referred to condition of **Low Wind Shear**

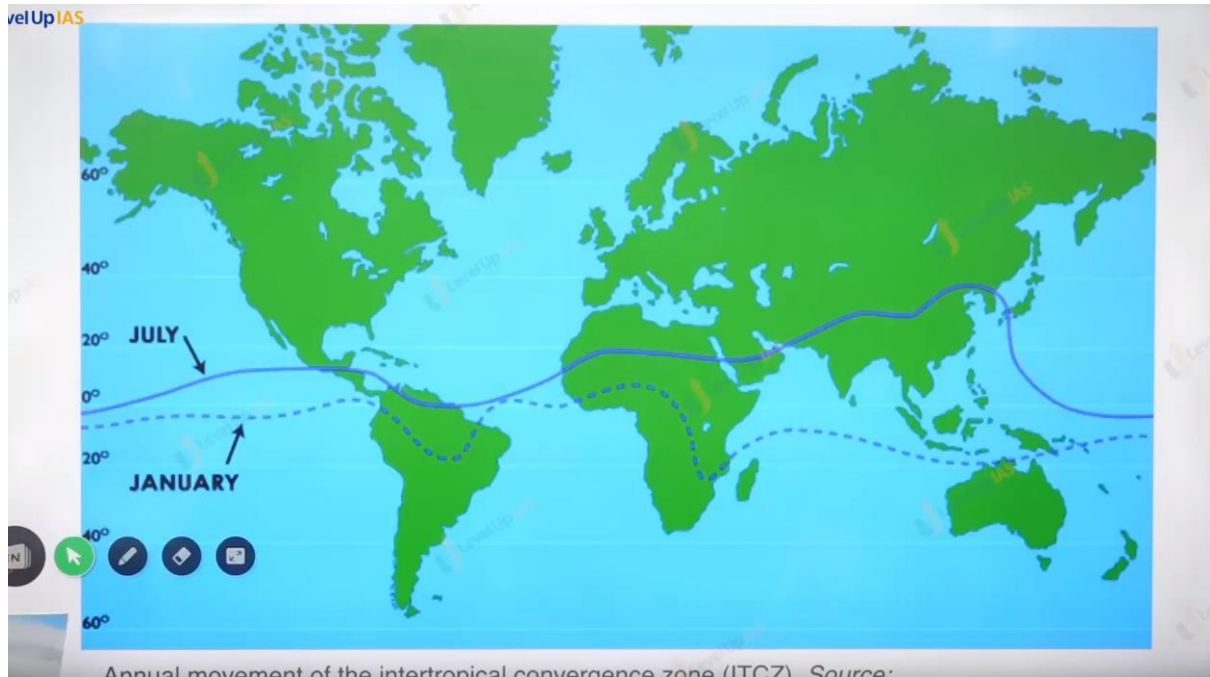
How tropical storms are formed
High humidity and ocean temperatures of over 26°C are major contributing factors

Water evaporates from the ocean surface and comes into contact with a mass of cold air, forming clouds

A column of low pressure develops at the centre. Winds form around the column

As pressure in the central column (the eye) weakens, the speed of the wind around it increases

Tropical cyclones develop around inter-tropical



Origin of the tropical cyclone

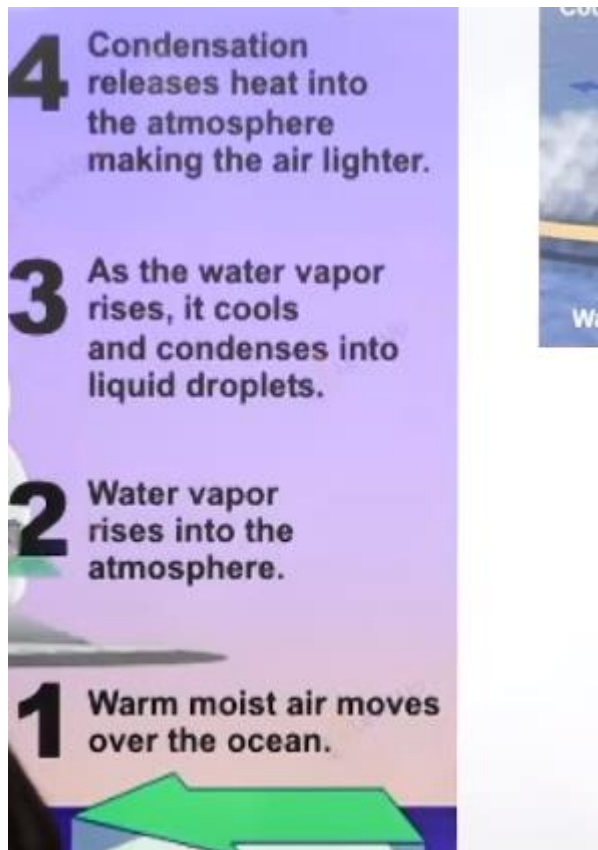
1. Tropical cyclone arises over the tropical ocean in the late summer and have a thermal origin (September to mid November).
2. Because of Coriolis effect convective current takes a whirlpool type motion.
3. The tropical cyclone formed and moved under the influence of trade winds towards the land.
- 4.

4 Condensation releases heat into the atmosphere making the air lighter.

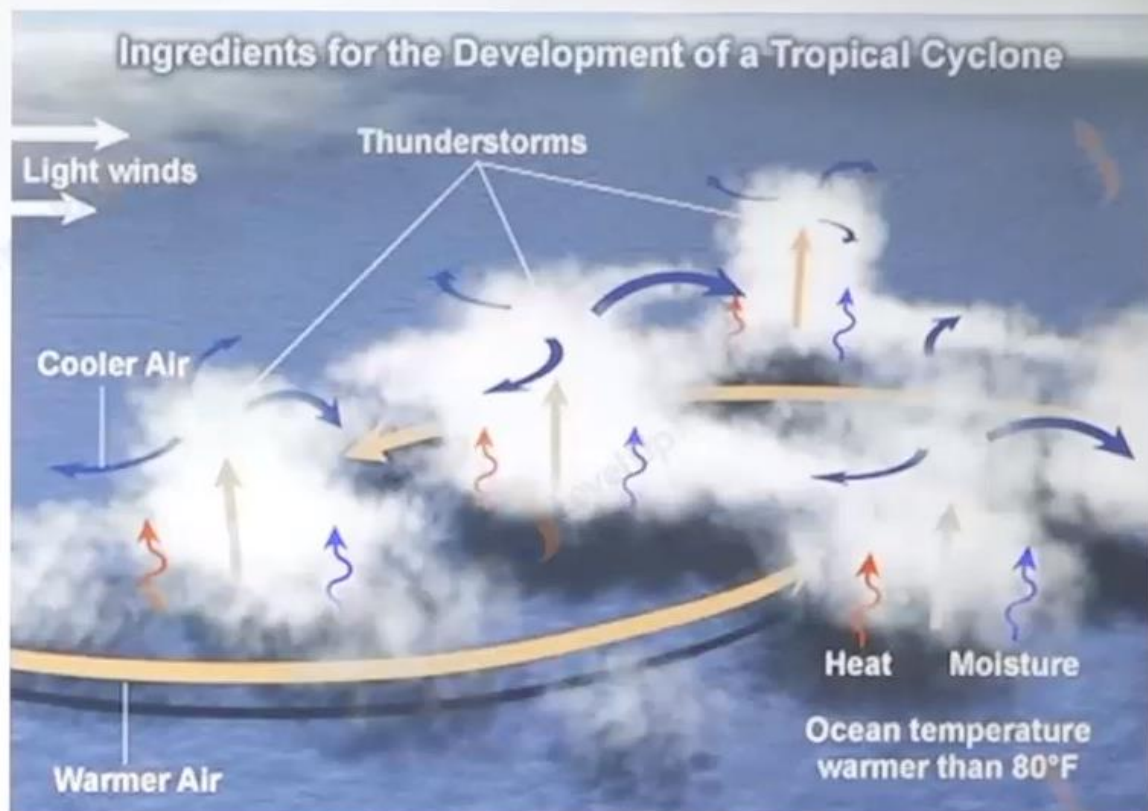
3 As the water vapor rises, it cools and condenses into liquid droplets.

2 Water vapor rises into the atmosphere.

1 Warm moist air moves over the ocean.



- 4** Condensation releases heat into the atmosphere making the air lighter.
 - 3** As the water vapor rises, it cools and condenses into liquid droplets.
 - 2** Water vapor rises into the atmosphere.
 - 1** Warm moist air moves over the ocean.
- 



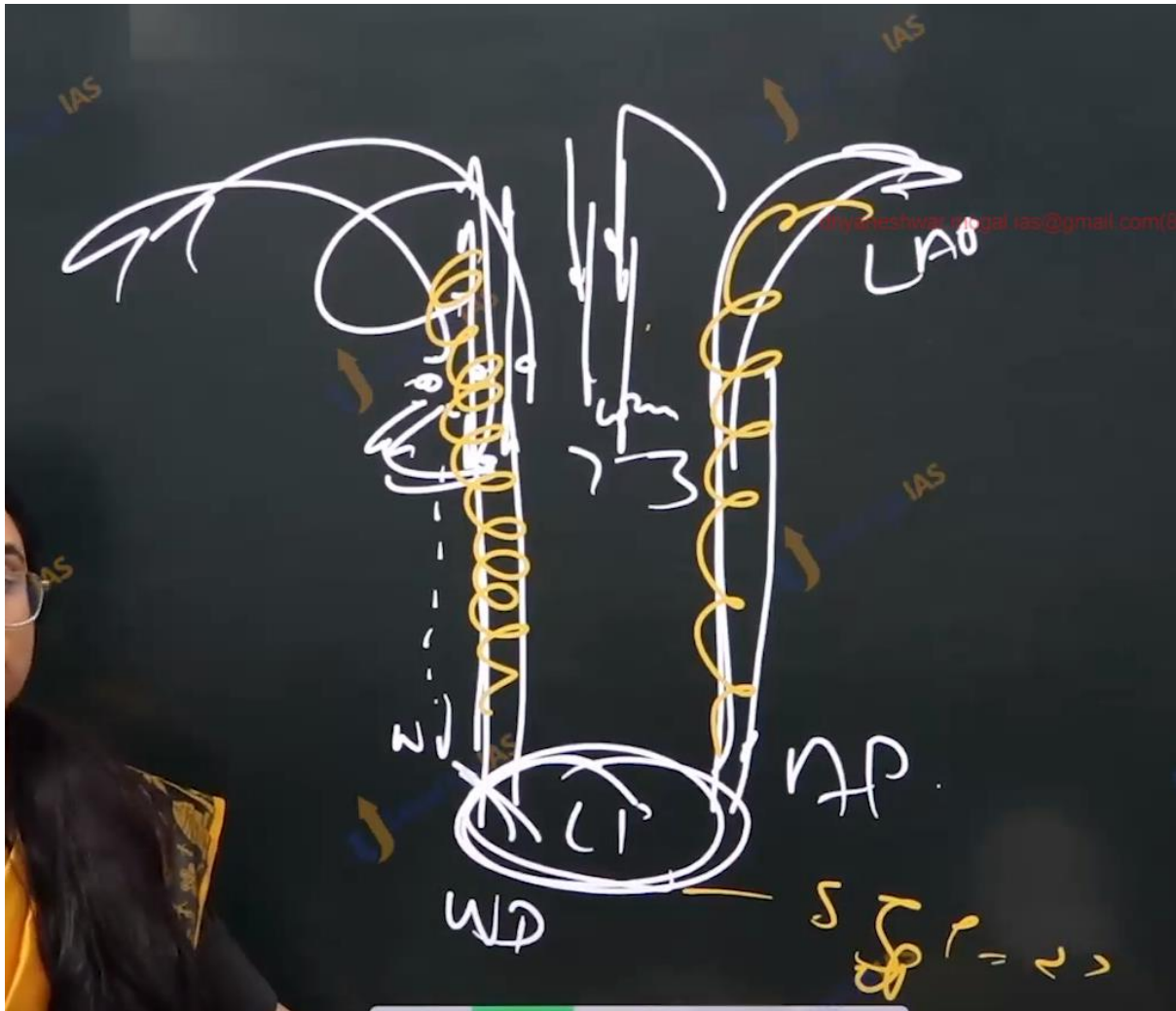
Stages of formation and initial development:

-
1. Transport of water vapour and heat from the warm ocean to the overlying air largely through evaporation.

 - a. From the sea surface, this leads to the formation of cyclonic storm.
 - b. Condensation of rising air releasing the heat and keeps the entire column warm.

 2. Mature stage:

 - a. Rising air intensifies as the tropical storm spreads horizontally at the tropopause. When the air spreads some portion of air migrates downwards and subsides in the eye.
 - b. When the air experiences subsidence the air is compressed which increases the temperature. This leads to the formation of a warm eye.
 - c. The sinking air in the eye does not reach the ocean surface but reaches only up to a depth of 1 to 3 km.



3. End of cyclone :

- a. As soon as the source of warm moist air abruptly cutoff tropical cyclone become weakens, on reaching the land moisture supplies is cutoff .
- b. The place where the tropical cyclone crosses the coast it called as land fall of cyclone.

Developmental Stages of Tropical Cyclone

3. End of cyclone

- As soon as the source of warm moist air is abruptly cut off, a tropical cyclone begins to weaken.
- On reaching the land the moisture supply is cut off and the storm dissipates.
- The place where a tropical cyclone crosses the coast is called the landfall of the cyclone.



Structure of tropical cyclone:

Structure of Tropical Cyclone: Eye, Eye Wall and Rain Bands

3. Spiral Bands:

- Curved bands of clouds and thunderstorms that go away from the eye wall in a spiral fashion.
- These bands are capable of producing heavy bursts of rain and wind.
- Outside the eye wall, series of rain bands forms and series of cumulus and cumulonimbus clouds drift into the outer region.
- Spiral bands are so named because they appear to spiral along the cyclone

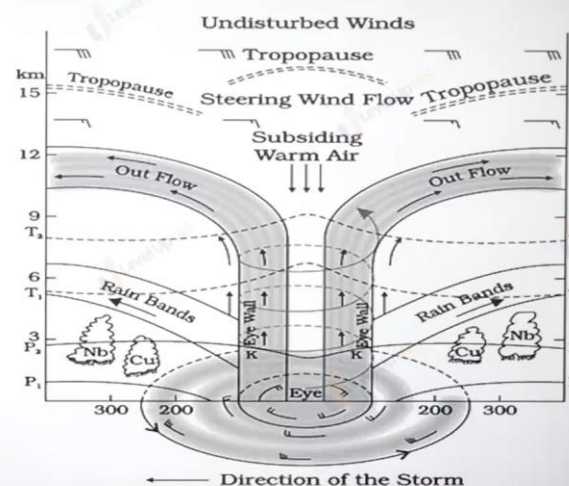


Fig: Vertical section of the tropical cyclone

Eye

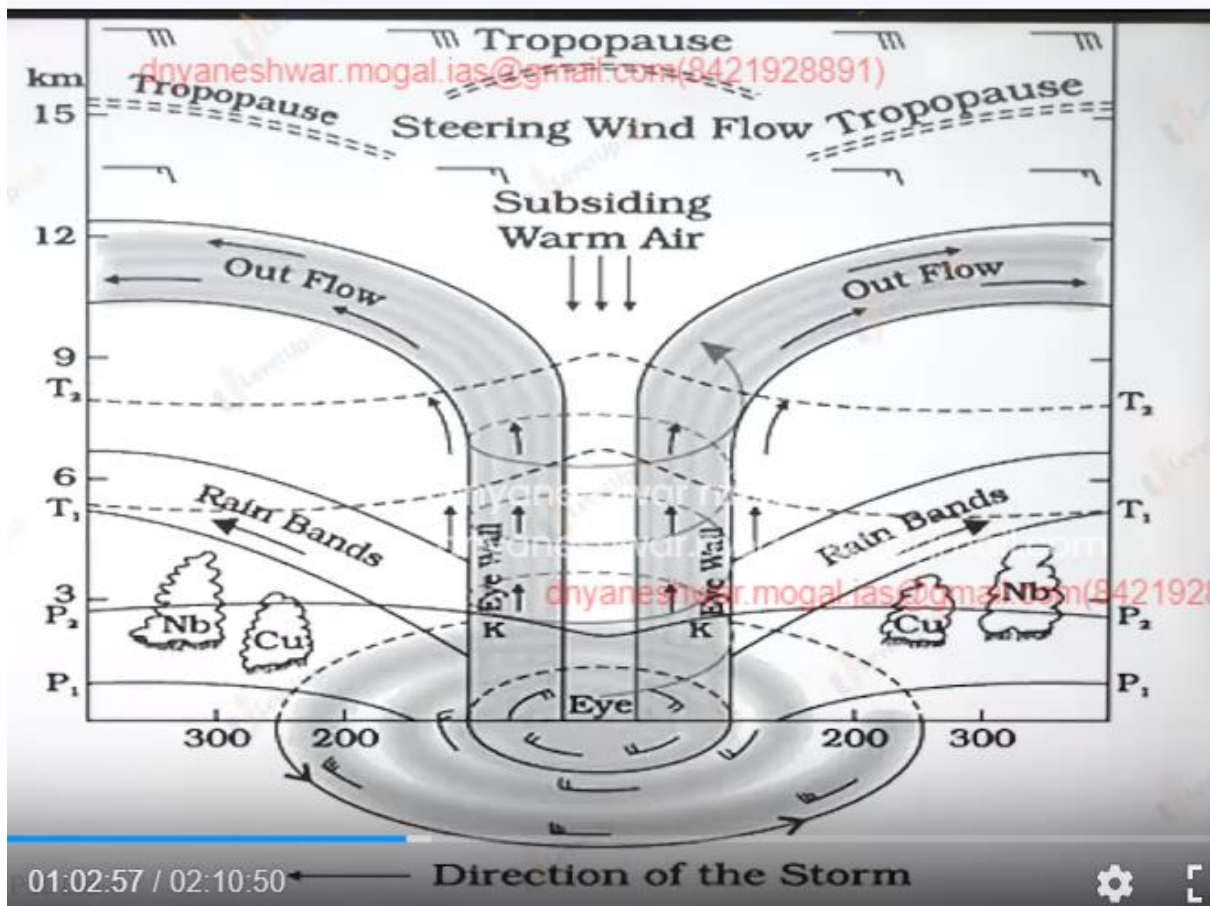
1. Eye is roughly circular area at the center of tropical cyclone with comparatively mild winds
 2. There is little to no precipitation in the eye.
 3. Eye is the area of lowest pressure zone. And with the warmest temperature.
 - 4.
-

Eye wall

1. Eye wall is roughly a circular area that surrounds the eye with spiraling and rising air.
 2. Eye wall has the strongest surface winds. The winds reach the velocity of 250 km per hour.
 3. Torrential rainfall occurs
-

Spiral bands:

1. The curly bands of clouds are moved along with the cyclone.
 2. They are also called rain bands.
-



Weeding ground and the local name:

Weeding ground are Caribbean sea, way of Bengal , Arabian sea towards north of Australia, south China sea.

1. Southeast caribbean region
2. Phillipn island
3. Sough china sea
4. Way of bangal and arabian sea
5. Towards madagaskar also north west of austrilia
- 6.

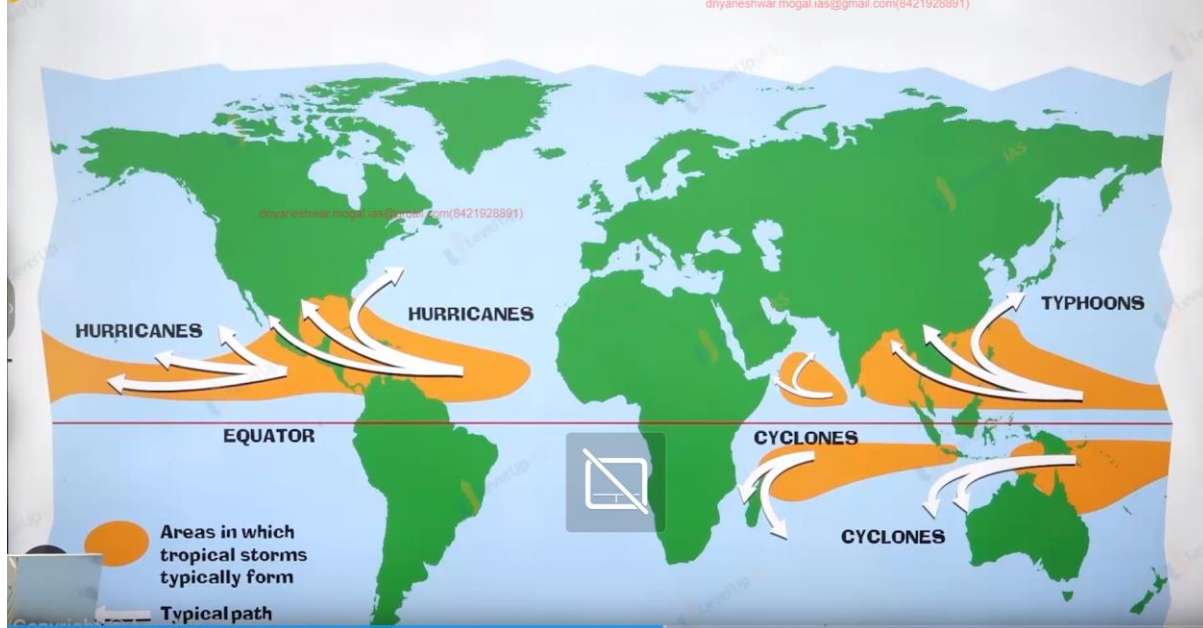
Favourite Breeding Grounds and Local Name

Favourite Breeding Ground:

- South-east Caribbean region (called **hurricanes**).
- Philippines islands, eastern China (called **typhoons**).
- Bay of Bengal and Arabian Sea (called **cyclones**).
- Around the south-east African coast and Madagascar-Mauritius islands.
- North-west Australia.

Local Names of Tropical Cyclone: In different areas, cyclones are referred to by different names.

1. Hurricanes – In the Atlantic and Eastern Pacific.
2. Typhoons – In Southeast Asia
3. Cyclone – In the Indian Ocean and Western Pacific around Australia



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2020

Consider the following statements:

1. Jet streams occur in the Northern Hemisphere **only**.
2. Only some cyclones develop an eye.
3. The temperature inside the eye of a cyclone is nearly 10 C lesser than that of the surroundings.

Which of the statements given above is/are correct:

- (a) 1 only (b) 2 and 3 only
(c) 2 only (d) 1 and 3 only

Jet stream develop in the both hemisphere, it develop in the upper part of atmosphere in the stratosphere.

Extra-tropical cyclone may not have a eye, they are develop due to warm and cold fault of air mass.

Eye is warmest and low pressure site in the cyclone.

2015

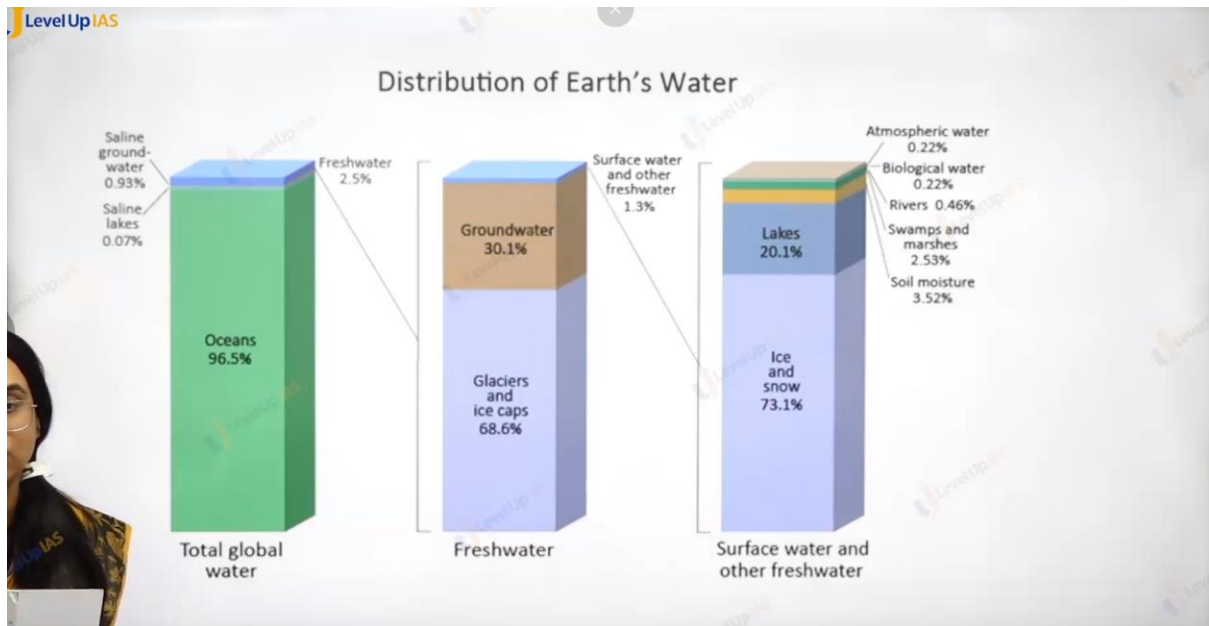
In the South Atlantic and South-Eastern Pacific regions in tropical latitudes, cyclone does not originate. What is the reason?

- (a) Sea surface temperatures are low
- (b) Inter-Tropical Convergence Zone seldom occurs
- (c) Coriolis force is too weak
- (d) Absence of land in those regions

Ans is b ITCZ mainly responsible for cyclone.

Cyclone mitigation and disaster preparedness:

1. Early warning system: tracking the path of tropical cyclone understanding the land fall, know the speed of cyclone etc.
 2. ISRO has done fairly good work as the casualty is reduced to nearly zero.
 3. Regulating the land leaves groves ex. Enforcing building code, coastal zone regulation, that is depending on the hazards prowess developmental activity should be modified.
 4. Engineering solution ex construction material should be cyclone material, water resistance.
 5. Shelter belt in the form of forest belt specially mangroves wetland to prevent coastal erosion
 6. Community participation: community is first responder.
 7. Color coding the cyclone: IMD (Indian meteorological department) issue the warning to alert warning from the natural hazards 4 color are used red, yellow green violet and orange.
 8. Violet means tropical cyclone can arrive in the next 6 hours meaning seek the shelter.
 9. Red means cyclone conditions are expected within 36 hours. It signifies the warning to protect oneself from cyclone
 10. Orange means tropical cyclone conditions are possible within 48-hour orange signifies preparing oneself from cyclone.
 11. Yellow signifies that the tropical cyclone crosses the threat and yellow means to be alert.
 12. Green means no hazards
-



Moisture in the atmosphere:

1. Distribution of earth's water

- a. 96.5 of the water in the oceans followed 1 percent of water in the saline lake and saline ground water. And 2.5 percent of water is fresh water.
- b. Out of 2.5 percent of water which fresh water. majority of water found in
 - i. Glaciers and ice caps
 - ii. Ground water
 - iii. Followed by lakes
 - iv. Water in the atmosphere
 - v. Followed by the water in the river.

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2013

On the planet earth, most of the freshwater exists as ice caps and glaciers. Out of the remaining freshwater, the largest proportion

(a) is found in atmosphere as moisture and clouds
(b) is found in freshwater lakes and rivers
(c) exists as groundwater
(d) exists as soil moisture



c. Ground water

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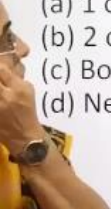
2021

With reference to the water on the planet Earth, consider the following statements:

1. The amount of water in the rivers and lakes is more than the amount of groundwater.
2. The amount of water in polar ice caps and glaciers is more than the amount of groundwater.

Which of the statements given above is/are correct?

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2



c. Only 2 , polar 70 and around 30 ground less 1% are in he river and lake atmosphere.

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Mains Questions

- Q) Briefly mention the alignment of major mountain ranges of the world and explain their impact on local weather conditions, with examples. 2021
- Q) How do the melting of the Arctic ice and glaciers of the Antarctic differently affect the weather patterns and human activities on the Earth? Explain. 2021
- Q) How does the cryosphere affect global climate? 2017
- Q) Discuss the concept of air mass and explain its role in macro-climatic changes. 2016
- Q) Tropical cyclones are largely confined to the South China Sea, Bay of Bengal, and the Gulf of Mexico. Why? 2014

2. Melting of arctic ice and Antarctic.

- They have the different effect on the weather changes and human activity.

4 air mass and role in the macro-climate changes.

Continental polar air mass , maritime tropic create cyclone.

5. condition cyclone itcz

Condensation, precipitation:

Condensation:

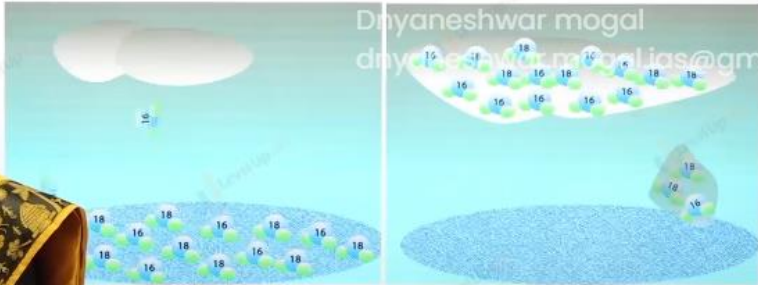
1. Conversion of water vapor to tiny water droplet as air raised it cools leading to saturation of air leading to condensation and precipitation.
2. Condensation can take place at or very near to the ground surface ex dew fog, mist. Frost(water to ice)
3. Condensation at the higher height it found the cloud.
- 4.

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Forms and types of condensation

Cooling → Saturation of air → Leads to condensation and precipitation

Condensation = water vapour converts into tiny droplets



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DEW:

1. It refers to the direct deposition of water vapor in the form of water droplets on the surface like grass plant, soil.

2. Conditions are

-
- a. Air above the ground in winter it become very cold and unable to hold the cold and deposits the water vapor on object
 - b. Conditions needed are clear sky, calm air, relatively high humidity, cold and long winter nights.
 - c. In dew condensation occurred above the freezing point.
 - d.
-

DEW

- It refers to direct deposition of water vapour in form of water droplets on surface like grass, plants and soil
- Conditions= Air above the ground in winters becomes very cold and is not able to hold water vapour and deposits the water vapour on objects
- In Dew, Condensation occurs above the freezing point
- The ideal conditions for its formation are clear sky, calm air, high relative humidity, and cold and long nights.



(23) Condensation, Precipitation

FROST

- Transformation of water vapour directly into solid form at ground surface
- The ideal conditions for the formation of white frost are the same as those for the formation of dew, except that the air temperature must be at or below the freezing point.



Frost:

-
1. Transformation of water vapor directly into solid form at the ground surface
 2. The ideal condition for frost are same that of dew except the temperature must be at below the freezing point.
-

FOG & MIST

-
1. The condensation of water vapor into water droplets on the condensation nuclei like dust particles, smoke, etc. suspended in air.
 2. These condensation nuclei have a tendency attract the water and therefore they are also called as high grow scope nuclei this suspended water droplets forms fog and mist.
 3. Fog and mist are suspended water droplets in the air and therefore they reduce the visibility.
 4. Mist has moister as compared to fog.
 5. Mist has microscopic water droplets as compared to fog.
-

Formation of fog



Water
Vapour



Cooling/
Condensation



Fog/ clouds

FOG AND MIST

- Condensation of water vapor into water droplet on condensation nuclei like dust particles, smoke etc suspended in air. (Air has dust particles, smoke, soot and these acts as condensation nuclei and hygroscopic nuclei (attracts water). These suspended water droplets forms fog/mist. Fog and Mist are suspended water droplets in the air and so reduces visibility. Difference between fog and mist is wrt size of water droplets. In mist water droplets are microscopic (v.v. small) as compared to fog which has more moisture as compared to mist.

FOG

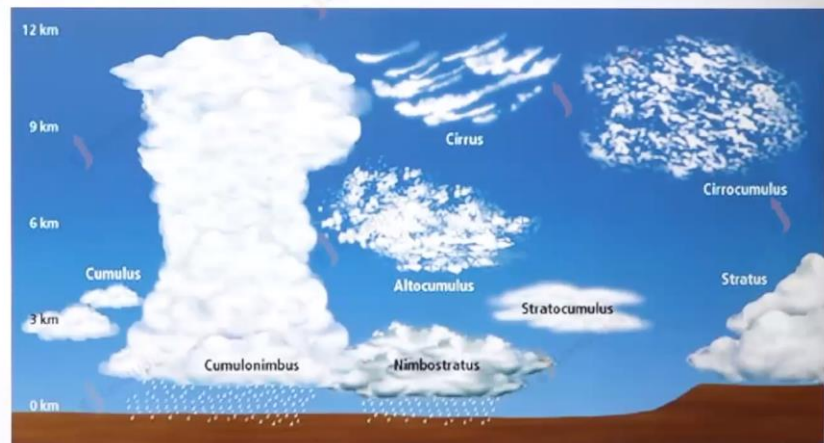


MIST



CLOUDS = Aggregate mixture of minute water droplet, ice particle or both in air at considerable elevations

- Condensation refers to conversion of water vapour to water droplet
- Precipitation= Falling down to water droplets/ice when the air is unable to hold it.



Cloud:

1. Here in the cloud Condensation take place at the higher height.
2. Cloud refers to aggregate mixture of minute water droplet, ice particles or both. found in the air at considerable elevations.
- 3.

Type of cloud:

1. Cirrus:

- a. high altitude clouds were found around 6-10 km altitude.
- b. They are thin cloud and white cloud.

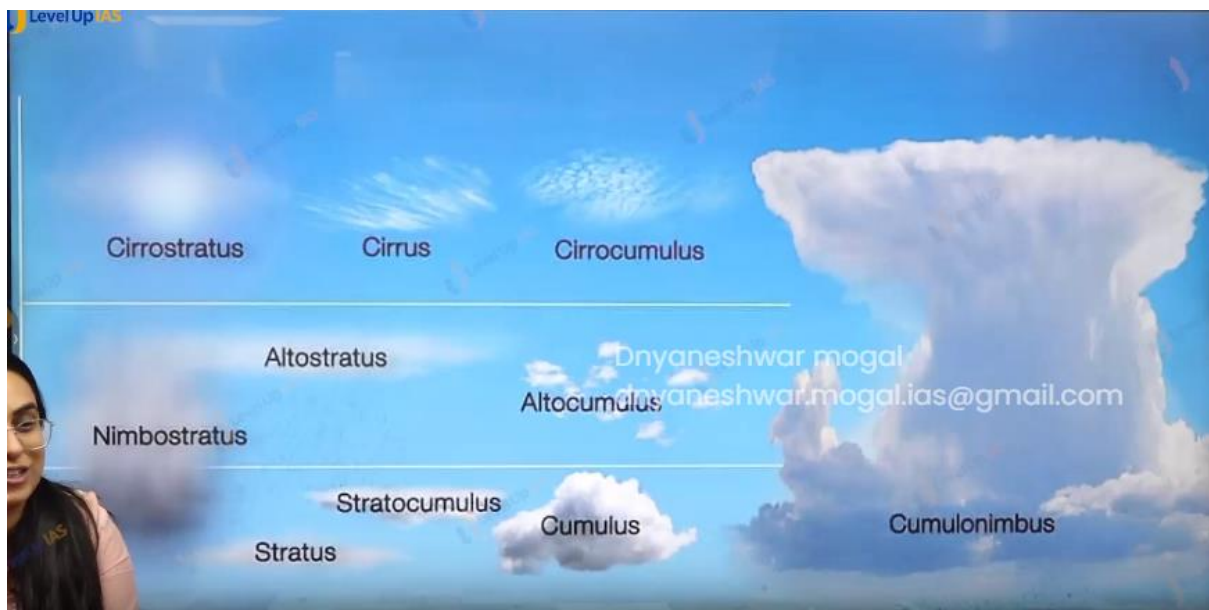
2. Alto:

- a. It is a medium level cloud. Found at an altitude of 5 km.
- b. Low level clouds are mainly stratus clouds.
- c. A cloud with high vertical clouds is cumulonimbus clouds

3. Cumulus clouds: cumulus means wooly or bumpy.

4. Stratus implies thin sheet.

5. Nimbus: Rain bearing,



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Types of clouds

- **High Clouds:** Cirrus, Cirrostratus, Cirrocumulus
- **Middle clouds:** Altostratus and Altocumulus
- **Low clouds:** Stratocumulus and Nimbostratus and

extensive vertical development – Cumulus and Cumulonimbus.

Type of cloud:

1. High cloud: cirrus cirrostratus, cirrocumulus.
2. Middle level cloud: altostratus, and altocumulus.
3. Low level cloud: stratocumulus and nimbostratus and stratus, and cumulus
4. The cloud very extensive vertical extends is cumulusnibus cloud will shape cloud.
5. Cirrus :

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- Copyr



- a. White colour cloud and indicate fair weather.
- b.
-
- Altostratus:
- a. It is white and woolly and bumpy cloud.
- b.
-
- Altostratus :
- a. It resembles thin white sheet in the sky. Sun and moon shines with hallow. Halo.

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High Clouds: Cirrus, Cirrostratus, Cirrocumulus

- **Cirrus:** White Coloured and indicate fair wather
- **Cirrocumulus:** White, woolly cloud
- **Cirrostratus:** Resembles thin white sheet in the sky. Sun and moon shines with a halo

CIRROSTRATUS

CIRROSTRATUS

CIRROCUMULUS

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4. Alto stratus :

- a. They are medium level dance cloud.

5. Altocumulus:

- a. It is white woolly bumpy cloud.
- b.

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Low Clouds: Stratus, StratoCumulus, Nimbostratus

- **Stratus:** Very Low cloud. It brings drizzle, Grey Coloured
- **Stratocumulus:** Low level Bumpy cloud.
- **Nimbo Stratus:** Dark Cloud. It is also known as Rain Cloud. Brings continuous Rain

STRATUS

STRATOCUMULUS

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NIMBOSTRATUS

6. Stratus :

- a. They are very low cloud. It brings drizzle(minute droplets), grey color.

7. Stratocumulus : low level bumpy cloud.

8. Nimbo status: dark cloud . it is also called as rain cloud and it bring continuous rainfall.

9.

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Clouds with great vertical extent: Cumulus, Cumulonimbus

- **Cumulus** = Vertical cloud, Rounded Top and Horizontal Base. Fair Weather Cloud. It is associated with rising convection current.
- **Cumulonimbus Cloud** = Overgrown cumulus cloud, Looks like an anvil. Brings Convectional Rain accompanied by lightening and thunder



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10. Cumulus:

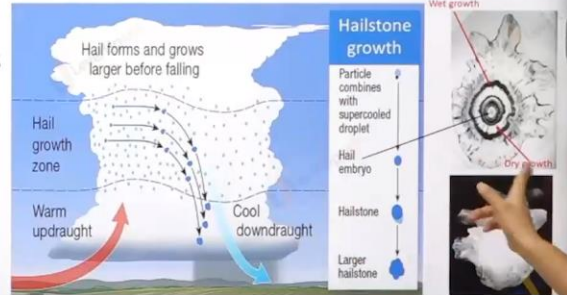
- i. Vertical cloud with rounded top and horizontal base. It is fair weather cloud and it is associated with raising conventional current.

11. Cumulonimbus cloud:

- i. It is overgrown cumulus cloud it looks like anvil and bring convectional rainfall accompanied by the thunder and lightning.

Types of Precipitation

- Condensed particles to grow in size and fall on Earth as precipitation. This may take place in liquid or solid form.
- Precipitation in the form of water is called **rainfall**
- Snowfall:** When the temperature is lower than zero degree Celsius, precipitation takes place in the form of fine flakes of snow.
- Sleet** is frozen raindrops and refrozen melted snow-water.
- Hailstone** = Hailstones have several concentric layers of ice one over the other. These are formed by the rainwater passing through the colder layers.



Types of rainfall

- Convective Rain:** The air on being heated, becomes light and rises up in convection currents. As it rises, it expands and loses heat and consequently, condensation takes place and cumulous clouds are formed. With thunder and lightening, heavy rainfall takes place but this does not last long. Such rain is common in the summer or in the hotter part of the day. It is very common in the equatorial regions

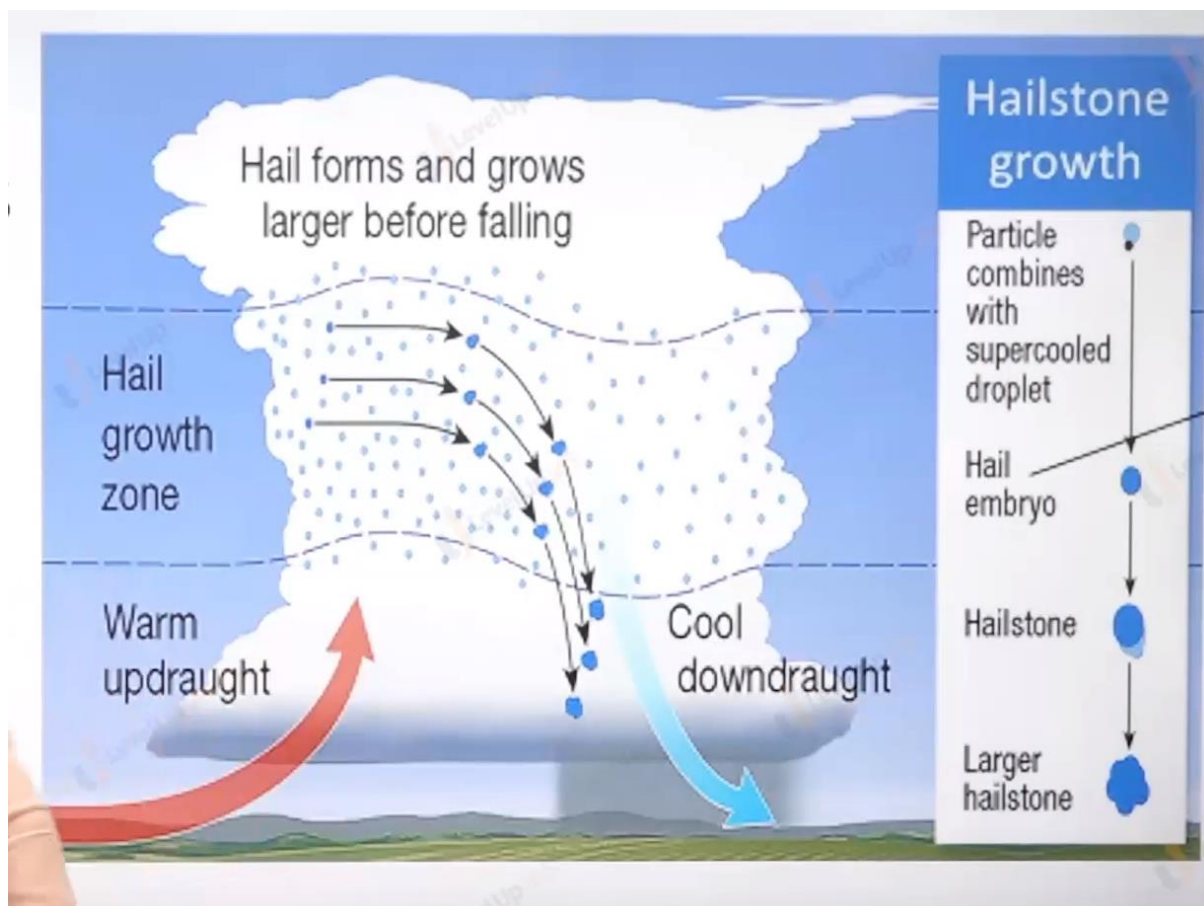


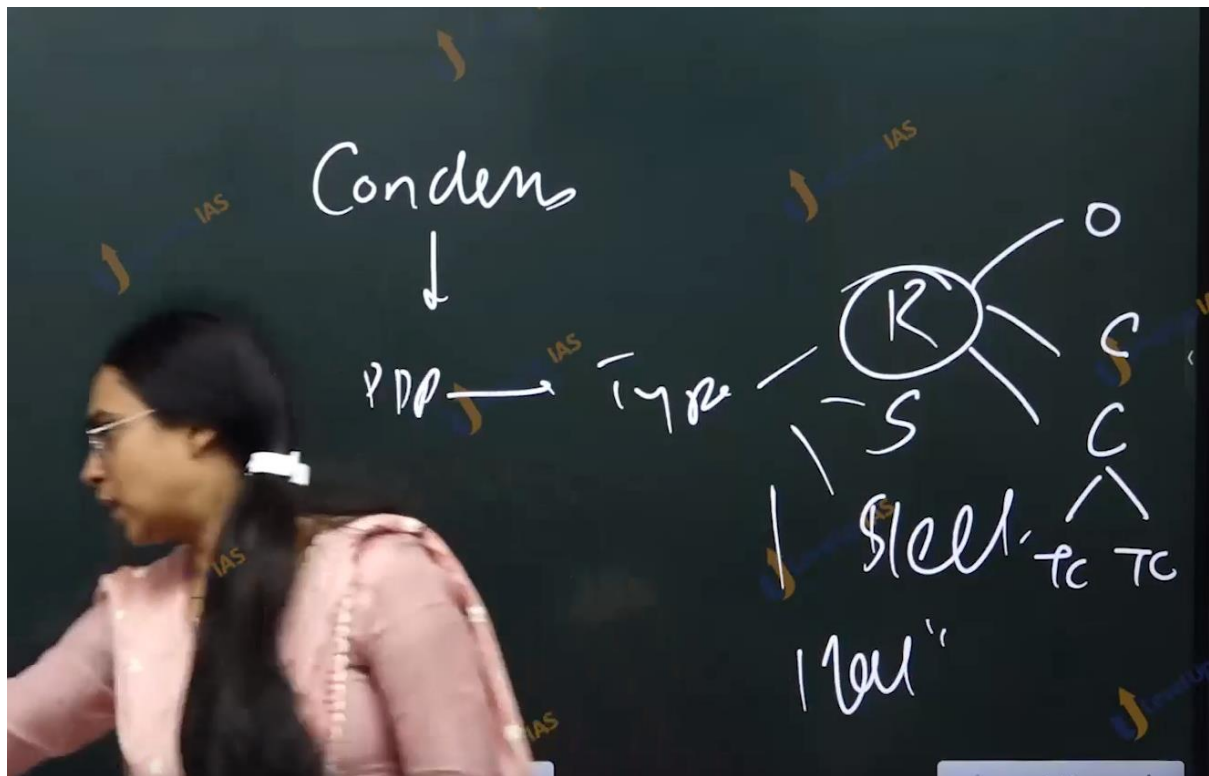
Precipitation:

1. Condensed particles can grow in size and fall on earth as precipitation.
2. This takes place in liquid or solid form
3. Precipitation in the form of water is called rainfall.
4. Snowfall:
 - a. When the precipitation takes place in the form of fine flakes of snow. That is temperature is less than 0 degrees Celsius.
5. Sleet: it is frozen raindrops that melt on the way and refreeze on coming in contact with cold air above the earth surface.

6. Hail stone:

- a. It has concentric layers of ice on over other.
 - b. The successive coating of ice is formed because water vapor experiences many updraughts and downdraughts in the cumulonimbus cloud.
-





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Types of rainfall

- Convectional Rain:** The air on being heated, becomes light and rises up in convection currents. As it rises, it expands and loses heat and consequently, condensation takes place and cumulous clouds are formed. With thunder and lightening, heavy rainfall takes place but this does not last long. Such rain is common in the summer or in the hotter part of the day. It is very common in the equatorial regions

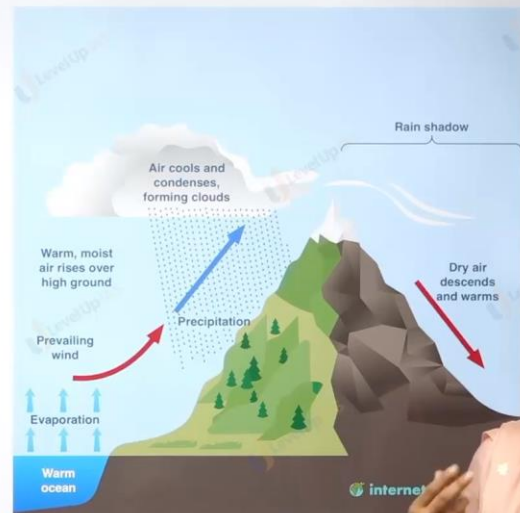
The sun heats the ground

The ground heats the air causing it to rise

The rising air cools & condenses

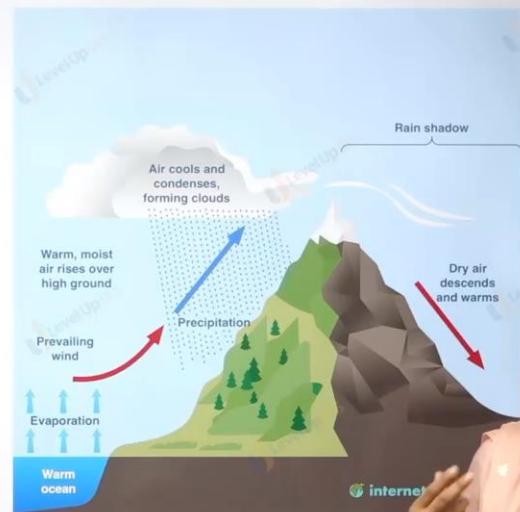
Types of rainfall

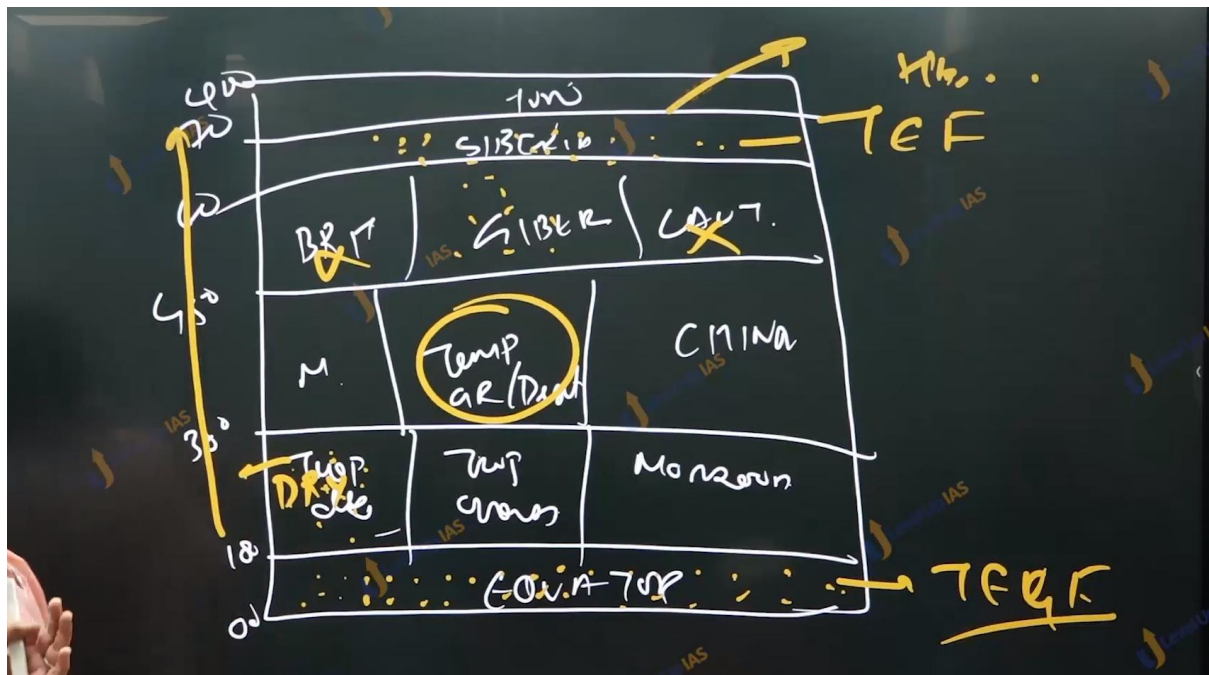
- **Orographic Rain/ Relief Rain:** Air comes across a mountain, is forced to ascend and as it rises, it expands; the temperature falls, and the moisture is condensed. Windward slopes receive greater rainfall. After giving rain on the windward side, winds on the leeward slope descend, their temperature rises and its capacity to take in moisture increases and hence, these leeward slopes remain rainless and dry. The area situated on the leeward side, which gets less rainfall is known as the rain-shadow area.
- **Cyclonic Rain:** Tropical Cyclone and Temperate Cyclone



Types of rainfall

- **Orographic Rain/ Relief Rain:** Air comes across a mountain, is forced to ascend and as it rises, it expands; the temperature falls, and the moisture is condensed. Windward slopes receive greater rainfall. After giving rain on the windward side, winds on the leeward slope descend, their temperature rises and its capacity to take in moisture increases and hence, these leeward slopes remain rainless and dry. The area situated on the leeward side, which gets less rainfall is known as the rain-shadow area.
- **Cyclonic Rain:** Tropical Cyclone and Temperate Cyclone





World Distribution of Rainfall

- **Equatorial Belt:** Convictional Rainfall, Very Heavy Rainfall. Average rainfall of 300 cm/year.
- **Tropical Monsoon:** 75-225 cm rainfall, Not a dry climate, Rainfall in monsoon month.
- **Tropical Grassland:** Summer rainfall, less than 75 cm, Savannah region
- **Desert:** Offshore trade winds, Rainfall less than 25 cm
- **(Warm Temperate) Mediterranean Climate:** Winter rainfall, <75 cm rainfall, semi arid region
- **Mid Latitude Desert:** dry condition and rainfall < 25 cm
- **Temperate Grassland:** Less than 50 cm/ year.
- **Cool Temperate:** Rainfall around 90 cm
- **Taiga Belt:** Rainfall throughout year, Rainfall well distributed. Coniferous forest. 70 cm
- **Polar Belt:** Rainfall less than 20 cm.

2013

"Climate is extreme, rainfall is scanty and the people used to be nomadic herders." The above statement best describes which of the following regions?

- (a) African Savannah
- (b) Central Asian Steppe
- (c) North American Prairie
- (d) Siberian Tundra

Ans b apply the answer. Try to eliminate option on the information.

2015

"Each day is more or less the same, the morning is clear and bright with a sea breeze; as the Sun climbs high in the sky, heat mounts up, dark clouds form, then rain comes with thunder and lightning. But rain is soon over.

"Which of the following regions is described in the above passage?

- (a) Savannah
- (b) Equatorial
- (c) Monsoon
- (d) Mediterranean

B. equatorial.

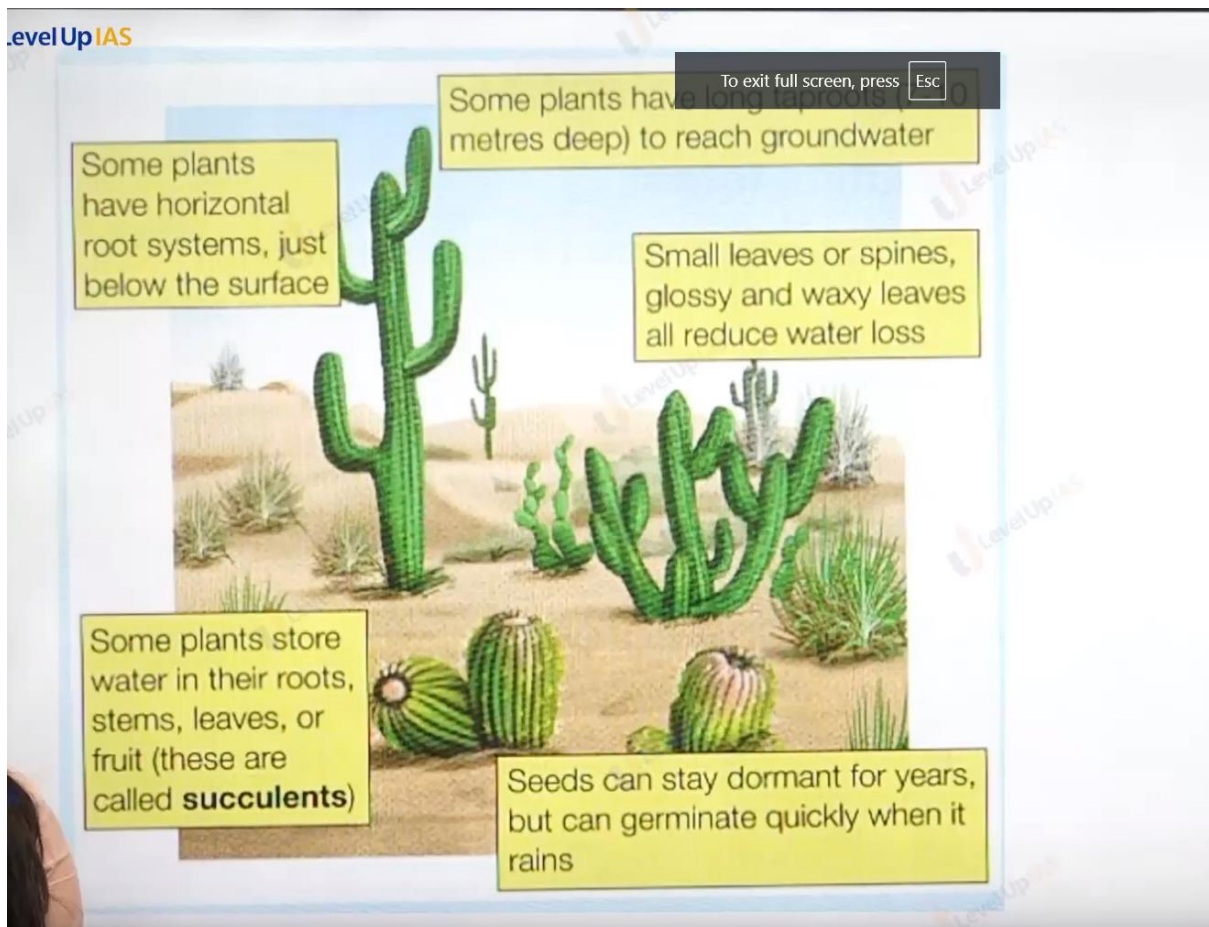
2018

Which of the following leaf modifications occur (s) in the desert areas to inhibit water loss?

1. Hard and waxy leaves
2. Tiny leaves
3. Thorns instead of leaves

Select the correct answer using the code given below:

- (a) 2 and 3 only
- (b) 2 only
- (c) 3 only
- (d) 1, 2 and 3



2013

Which of the following is/are unique characteristic/characteristics of equatorial forests?

1. Presence of tall, closely set trees with crowns forming a continuous canopy
2. Coexistence of a large number of species
3. Presence of numerous varieties of epiphytes

Select the correct answer using the code given below:

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Ans D.

2021

Leaf litter decomposes faster than in any other biome and as a result, the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up into the canopy vicariously, by climbing the trees or growing as epiphytes, rooted on the upper branches of trees." This is the most likely description of

- (a) Coniferous forest
- (b) Dry deciduous forest
- (c) Mangrove forest
- (d) Tropical rain forest

Answer:
